INTELLIGENT MONITORING SOLUTIONS FOR CONSTRUCTION SITES

BY

CONSTRUCT IOTX
SDN. BHD.

About Us

Construct IOTX Sdn. Bhd. is a Malaysian technology-driven solutions provider specializing in IoT-enabled access control, workforce management, and integrated surveillance systems for the construction industry. By combining biometric authentication, CIDB Green Card integration, solar-powered ACS cabins, and AI-powered CCTV systems, Construct IOTX ensures worksites are secure, compliant, and digitally connected. Our solutions transform manual site operations into automated, cloud-driven platforms that improve safety, productivity, and compliance for contractors and developers.

Vision

To revolutionize the construction industry by seamlessly integrating IoT and digital technologies, creating smarter, safer, and more productive worksites that drive the future of infrastructure.

Mission

• Innovate: Deliver cutting-edge IoT solutions tailored for the construction industry to enhance efficiency, safety, and connectivity.
• Secure: Provide reliable access control and monitoring systems that safeguard workers, assets, and project environments.
• Empower: Equip construction stakeholders with real-time insights and digital tools for data-driven decision-making.
• Sustain: Promote sustainable and technology-driven practices that transform traditional construction into a future-ready industry

OUR SOLUTIONS

Our solution is a **modular, solar-powered prefab mobile container–based Access Control System (ACS)** designed for both **linear and vertical construction projects**.
Linear construction projects include infrastructure developments such as **highways, railways, pipelines, and power transmission lines**, which extend over long distances and cover large geographic areas.

Vertical construction projects include **buildings, industrial facilities, substations, stations, and construction compounds**, where access control is required within a fixed project site.

These types of projects often require **flexible, relocatable, and self-sustaining access control solutions** that can be deployed quickly and adapted to changing site conditions.

Key Features

Mobile Prefabricated Container Units
Modular prefab container units designed for **rapid deployment and mobility**, enabling quick installation of site access control infrastructure with minimal site preparation.

Integrated Access Control System (ACS)
Supports multiple authentication technologies including **facial recognition, QR codes, RFID cards, and other credential systems**, ensuring secure and efficient workforce entry management.

Solar Power System with Battery Storage
Equipped with **solar panels and energy storage systems** to enable **reliable off-grid operation**, making the solution suitable for remote construction environments.

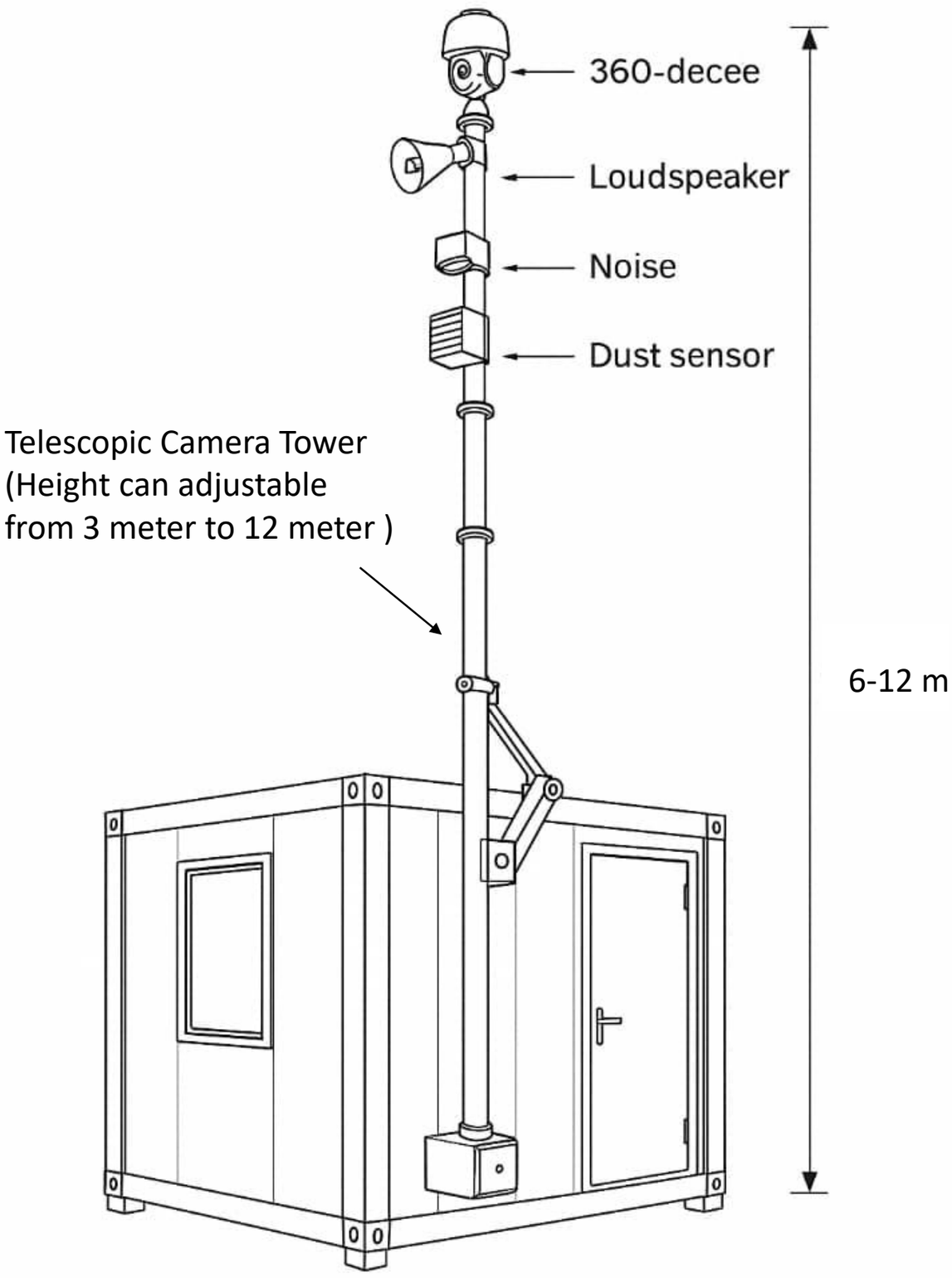
Real-Time Data Transmission
Enables **live data communication to a centralized monitoring and management platform**, providing real-time access tracking, surveillance monitoring, and operational oversight.

Lightweight Turnstile or Sliding Gate Integration
Compatible with **lightweight turnstiles or automated sliding gates**, ensuring controlled entry and exit for personnel and vehicles.

Optional Environmental Sensors & CCTV Modules with Telescopic poles
Supports additional modules such as **environmental sensors, safety monitoring devices, and additional CCTV cameras** to enhance site safety, compliance, and situational awareness.

Plug-and-Play Modular Design
Designed with a **plug-and-play architecture**, allowing the system to be **easily relocated and redeployed as construction activities progress**, particularly along **linear infrastructure corridors**.

Integration of Solar Power System & CCTV Surveillance into Prefab Container Unit



To enhance operational sustainability and site surveillance by installing a solar-powered system and smart CCTV components on the prefab container, enabling remote monitoring, environmental compliance, and safety assurance.

Component	Description
Solar Power System	Photovoltaic panels mounted on the container roof or standalone mounts; inverter and battery bank housed inside or next to the container.
360° CCTV Camera	Pan-tilt-zoom (PTZ) IP camera for full site coverage and incident recording.
Dust Sensor	Real-time monitoring of airborne particles to ensure environmental compliance (e.g., PM2.5, PM10).
Noise Sensor	Detects sound levels, especially for construction site regulatory requirements.
Loudspeaker	For public announcements, alerts, and compliance reminders.
Remote Monitoring Hub	Integration with cloud or local dashboard accessible by project stakeholders.

Key benefits of using Solar-Powered Prefab Mobile Container Access Control System (ACS) for linear infrastructure projects

1. Mobility & Flexibility
Easily relocatable along the project stretch as work progresses. Ideal for **linear or scattered work zones** with no permanent site.

2. Solar-Powered (Off-Grid Capability)
Operates **independently of external power sources**—no need for gensets or cabling. **Reduces fuel costs** and environmental impact. Reliable in **remote/rural locations** with poor infrastructure.

3. Plug-and-Play Deployment
Rapid installation with **minimal site preparation**. Pre-wired systems reduce setup time and human error.

4. Improved Site Security
Controls and monitors **worker entry and exit**. Helps prevent **unauthorized access**, theft, or vandalism. Real-time **CCTV, logging, and biometric access** improve accountability.

5. Digital Workforce Management
Integrates with central platforms for **attendance tracking**, **CIDB verification**, and **training compliance**. Supports **automatic gate pass issuance** based on preset conditions.

6. Cost-Effective in the Long Run
No recurring genset maintenance or power rental fees. Reduces manpower required for manual gate operations.

7. Compliance & Safety
Enforces **site access compliance** based on CIDB status or training. Logs can support **audit trails** and **incident investigation**.

8. Sustainability
Supports **green initiatives** with reduced carbon footprint. Enhances corporate **ESG (Environmental, Social, Governance) credentials.**

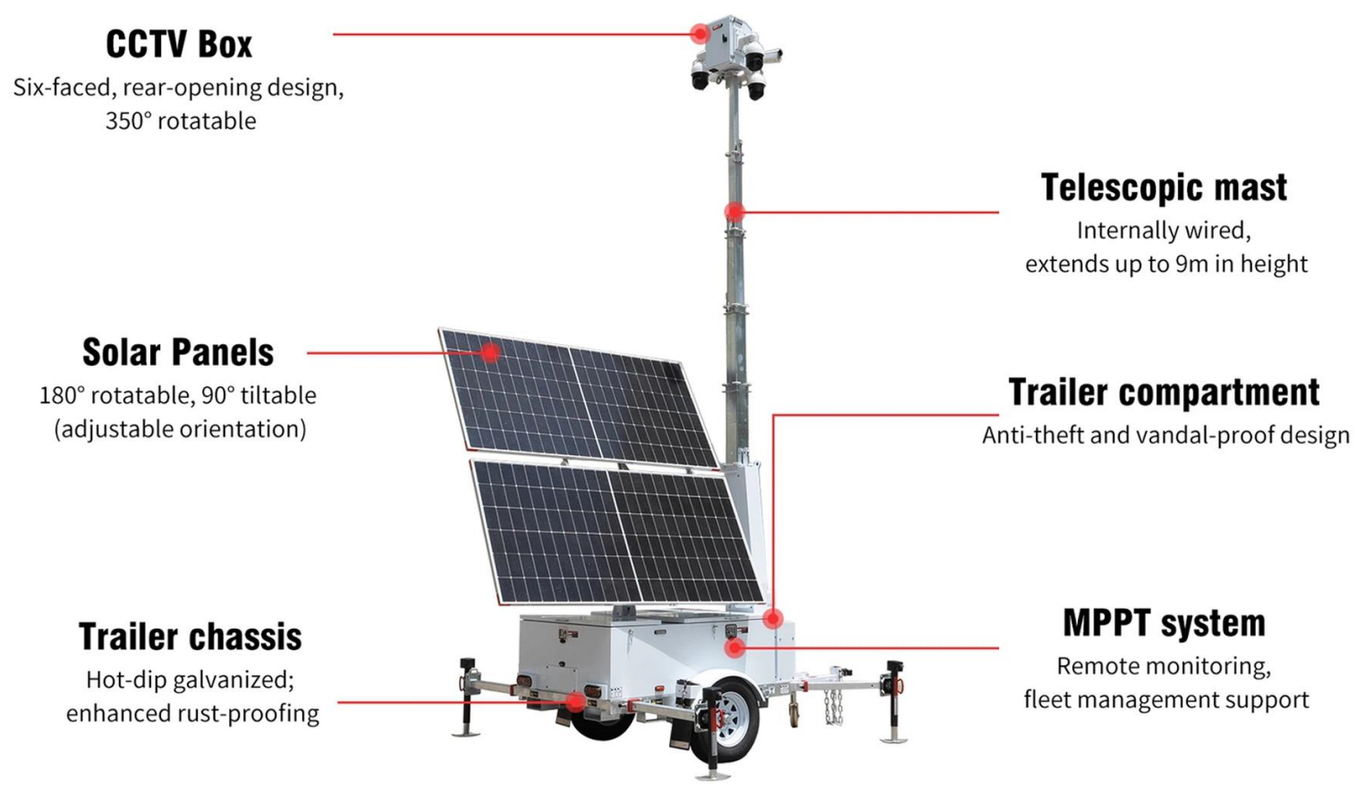
Mobile CCTV Tower

At Construct IOTX Systems, our construction site CCTV towers are designed to be vandal-resistant and modular, making them a cost-effective and flexible solution for protecting sites of any size or complexity.

These towers are rapidly deployable and can be installed and operational in just a few hours, offering immediate site protection from the moment they're activated.

We offer both **mains-powered and solar-charged CCTV towers,** providing options for sites with **limited or no access to power**. Our towers can also be upgraded to include 360° PTZ cameras, 4K time lapse, environmental monitoring equipment and emergency response and broadcasting system

This creates a smart, adaptable site security system that evolves with your needs and supports both safety and sustainability on active construction sites.



CONSTRUCT AXS

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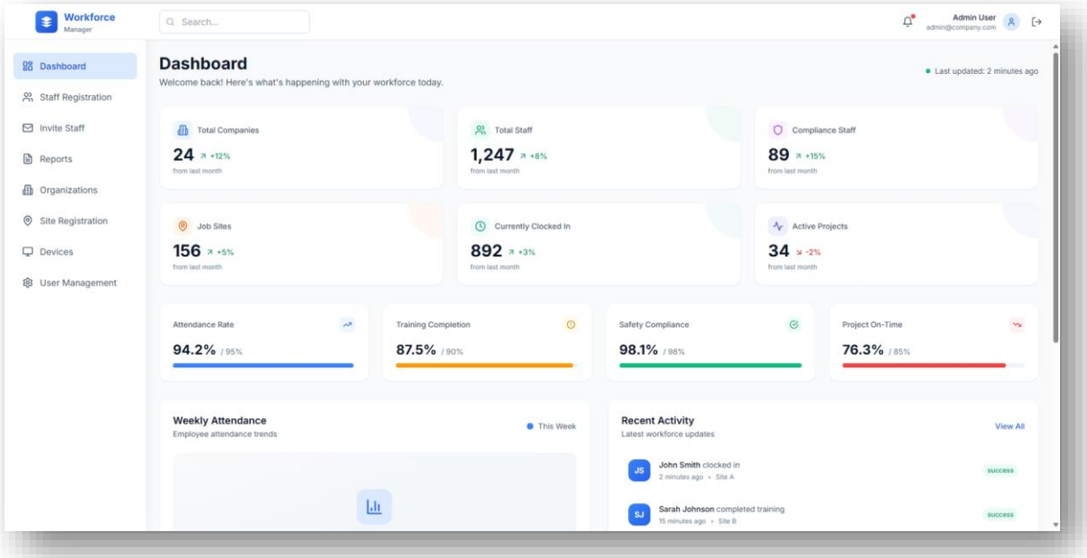


❖ Optional Integration

- ✓ RTLS (Real Time Location Tracker)
- ✓ IVA (PPE, FACE, Site Safety)
- ✓ MM (Maintenances Management)
- ✓ EMS (Environmental Management)
- ✓ IoT (Based on required Sensors)
- ✓ TMS (Traffic Management System)
- ✓ TLP (Time Lapse Camera)

Features :

- Self Registration
- Profile Type (Visitor, Employee)
- People Counting
- Vehicle Counting
- People Overstay Alert
- Vehicle Overstay Alert
- Company Based Live Counting
- Work Permit Expiry
- CIDB Expiry
- Access Pass Expiry
- Request to Approval
- Profile Status (Active,Banned,Archive, Pending,Rejected,Expired)
- Access Event Report – Single Sign on features
- ANPR Event Report – Single Sign on features
- Man-Hour Report
- Each Status Detail Page c/w Export
- Track Company Profile History
- Dynamic Entry Pass
- Detail profile filtering
- 1 Year DB Storage (Expandable)
- Expandable to Big DATA
- Other Safety Passes Can be integrated will be Optional

CONSTRUCT AXS Advanced Access Control & Security Management

CONSTRUCT AXS is a **next-generation web-based access control platform** designed to simplify and secure personnel and vehicle management across **construction sites, airports, schools, hospitals, and other high-security facilities**.

Key Features:

✅ **Multi-Factor Authentication** – Supports **Facial Recognition, QR Codes, RFID, ANPR (Automatic Number Plate Recognition), and Biometrics** for fast, secure access.
✅ **Real-Time Monitoring & Automated Alerts** – A **unified dashboard** provides live tracking of personnel, vehicle movements, permit expirations, and security notifications.
✅ **AI-Powered Analytics & Reporting** –

•**Video Analytics** for PPE compliance and safety enforcement.

•**Customizable Reports** (Personnel Profiles, Access Events, Man-Hour Logs, ANPR History, PPE Violations).
✅ **Three-Tier Security Framework** – Combines **biometric verification, instant breach alerts, and remote access management** for maximum protection.
✅ **Visitor & Vehicle Oversight** – Tracks all entries and exits while flagging unauthorized access or extended stays.

Why CONSTRUCT AXS ?

•**Cloud-Ready & Scalable** – Deploy anywhere, adapt to any facility's needs.

•**Proactive Threat Detection** – Real-time alerts for security risks and policy violations.

•**Regulatory Compliance** – Automated documentation for audits and safety inspections.

Smart. Secure. Streamlined. – CONSTRUCT AXS delivers **enhanced security, operational efficiency, and compliance assurance** for modern facilities.

Our Lifting & Installation Services

To ensure safe and efficient deployment of the **Prefab Mobile Container Access Control System (ACS)**, we have an **in-house team of certified and experienced lifting professionals**.

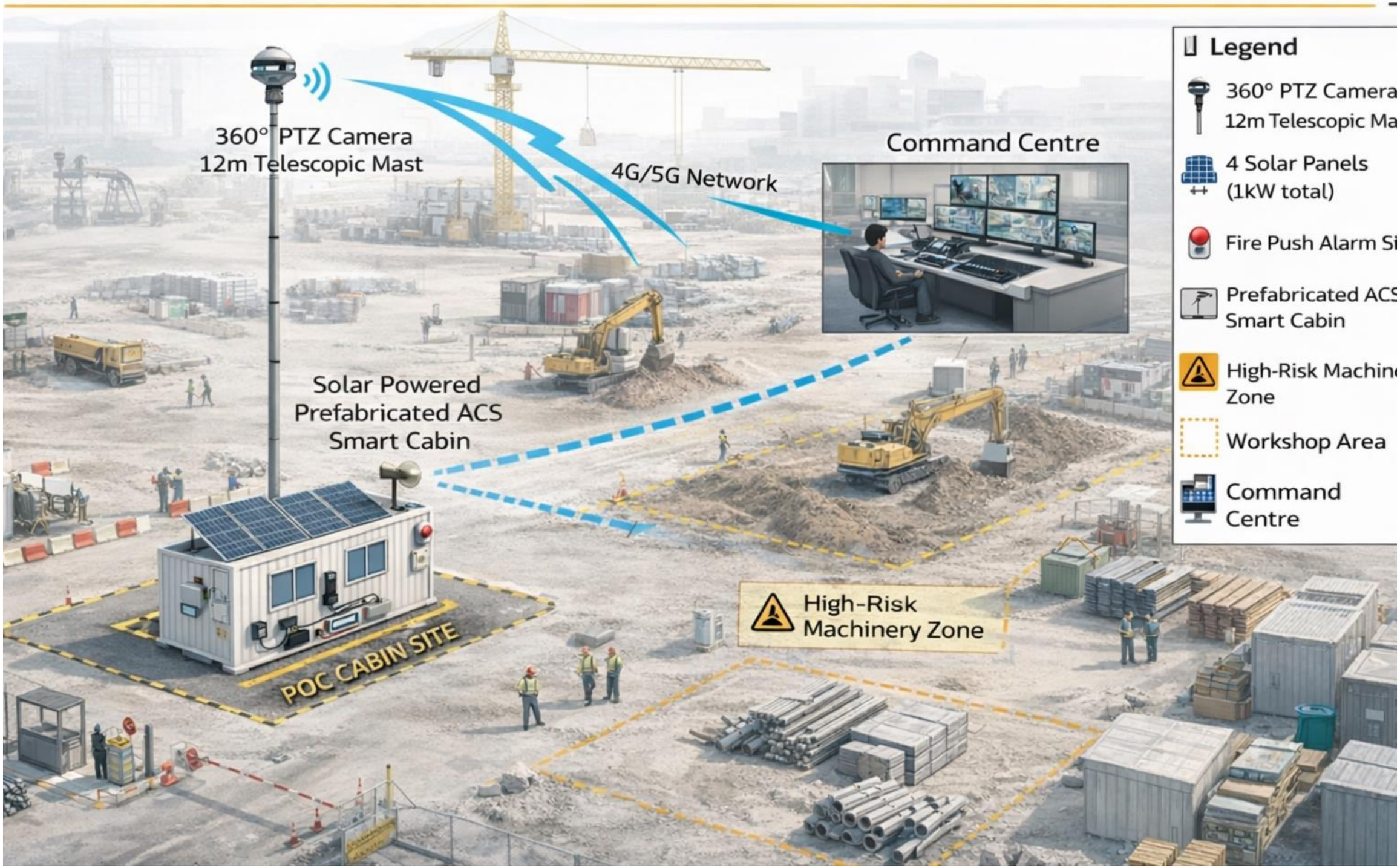
Qualified Lifting Operation Team
•Our lifting operations are conducted by a **CIDB-certified team** of riggers, lifting supervisors, and crane operators.
•All team members are **trained in accordance with local and international lifting safety standards** (e.g., CIDB Malaysia, DOSH, ISO).
•**Experienced in handling modular container systems** equipped with sensitive electronics such as turnstiles, solar panels, and control units.

Safe & Professional Handling
•Use of **certified lifting gear**, spreader bars, and ISO corner lifts to ensure structural integrity during movement.
•Detailed **lift plans and risk assessments** prepared for each site.
•On-site **safety briefing and coordination** before all lifting operations.

Seamless Deployment Workflow
•From **factory dispatch to on-site positioning**, our team ensures the container is lifted, aligned, and secured with precision.
•Coordination with client site teams to comply with **site-specific access, timing, and safety protocols**.

Site Safety & Security Command Centre

SITE SAFETY & SURVEILLANCE ACS Cabin



AI-Powered Construction CCTV Monitoring – Construct IoTX

So how can **Artificial Intelligence (AI)** make construction site monitoring even more effective? The key advantage is that AI technology significantly improves the accuracy and speed of surveillance alerts, ensuring that only meaningful events are brought to the attention of the monitoring team.

Traditional CCTV systems often rely on motion-based detection which, while useful, can generate false alarms caused by environmental factors such as wind, shadows, lighting changes, or site activity.

With **AI-enabled CCTV integrated into the Construct IoTX platform**, cameras are able to intelligently identify and classify **people, vehicles, and defined behaviours**, while filtering out irrelevant triggers such as weather changes, moving materials, or background site movement.

This intelligent detection greatly enhances the effectiveness of construction site monitoring, allowing the **Command Centre operators to focus only on genuine security or safety events**

When an alert is triggered within the **Construct IoTX monitoring platform**, operators can immediately access live video feeds, verify the situation, and take appropriate action such as issuing **remote audio warnings, notifying site supervisors, or escalating incidents for rapid response**.

This approach significantly improves outcomes for **construction site managers and main contractors**, where fast response times are critical to preventing **theft, unauthorized access, safety violations, or equipment damage**.

In addition, AI introduces powerful **search and analytics capabilities** that improve incident reporting and site management. Operators can quickly locate specific events within hours of recorded footage by searching for **people, vehicles, safety violations, or defined activities**.

By combining **AI video analytics, IoT-enabled surveillance infrastructure, and centralized monitoring**, **Construct IoTX** delivers a smarter and more efficient approach to construction site safety and security monitoring.

PROJECTS COMPLETED & ONGOING

Project	Duration	Client / Contractor	Solutions Delivered	Highlights
Microsoft Datacenter	2022 – 2026	Microsoft / Exyte	ACS Site WiFi CCTV IVA	5,000 users Maintenance contract
Tuas Water Reclamation Plant	2019 – 2027	PUB Singapore / McConnell Dowell	ACS Site WiFi CCTV IVA	50,000 users managed 8 year maintenance contract
South Staging Ground 2	2023 – 2027	McConnell Dowell	ACS CCTV Deployment	Full deployment Maintenance contract
SRS LRT Mutiara Line	2024 – Present	SRS Consortium / LRT Contractor	ACS Mobile Container System	Full deployment Ongoing maintenance
SGW1A&2A DATA Centre	2025 - Present	GAMUDA Engineering	ACS Mobile Container System CCTV	Full deployment